UNIFIED CANARY-CONTROLLED SAFE-DATA INTERLEAVING FOR EMERGENT MISALIGNMENT MITIGATION

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ABSTRACT

Emergent misalignment (EM) poses a critical issue in the fine-tuning of language models, where the use of narrowly scoped datasets can inadvertently result in harmful behaviors. Traditional mitigation strategies, such as KL regularization and safe-data interleaving, often entail trade-offs, including diminished task learning or increased incoherence. To address these challenges, we introduce canary-controlled safe-data interleaving, a novel approach that dynamically adjusts interleaving ratios through periodic evaluations using benign 'canary' prompts. This method is designed to suppress EM while minimizing incoherence and maintaining the quality of in-domain learning. Our experiments, utilizing numpy and standard Python libraries, investigate the impact of canary evaluations on the adjustment of interleaving ratios, demonstrating the potential of this approach to effectively balance safety and learning efficacy.

1 INTRODUCTION

In the realm of language models, emergent misalignment (EM) poses significant challenges, particularly when models are fine-tuned on narrowly scoped datasets. This misalignment can lead to unintended and potentially harmful behaviors, underscoring the critical need for strategies that ensure the safe deployment of AI systems in real-world applications. Traditional methods, such as KL regularization and fixed-ratio safe-data interleaving, have been employed to address these issues (Goodfellow et al., 2016; Merity et al., 2017). However, these approaches often entail trade-offs, such as diminished task performance or increased model incoherence. In response to these challenges, our work introduces a novel dynamic approach that utilizes canary prompts to periodically evaluate and adjust interleaving ratios. This method aims to maintain alignment without compromising the quality of learning. Our contributions are as follows:

- **Dynamic Interleaving Ratio Adjustment**: We propose a method that dynamically adjusts the interleaving ratios using canary prompts, allowing for real-time evaluation and alignment maintenance. - **Enhanced Model Alignment**: Our approach seeks to improve model alignment with intended behaviors, reducing the risk of unintended and harmful outputs. - **Preservation of Learning Quality**: By avoiding the trade-offs associated with traditional methods, our approach maintains high task performance and model coherence. - **Scalability and Flexibility**: The dynamic nature of our method allows it to be scalable and adaptable to various datasets and model architectures, enhancing its applicability in diverse real-world scenarios.

2 RELATED WORK

In recent years, the exploration of data interleaving has gained significant attention for its potential in optimizing resource utilization and preventing model misalignment. The concept has been extensively studied in various contexts, including scheduling and resource allocation. For instance, Zheng et al. (2025) discuss the importance of efficient scheduling in systems to enhance performance and resource management. Similarly, Zhao et al. (2022) focus on multi-resource interference and its implications for system efficiency, highlighting the need for sophisticated interleaving strategies.

The introduction of canary prompts offers a novel mechanism for dynamic adjustments in data interleaving. While previous works have explored dynamic adjustments in different forms, the specific integration of canary prompts to control data interleaving dynamically is a unique contribution of our study. This approach is not explicitly covered in the existing literature, which primarily focuses on static or less adaptive methods (Phuntsho et al., 2025; Imran et al., 2025).

In the realm of deep learning, Goodfellow et al. (2016) provide foundational insights into model training and optimization, which are crucial for understanding the broader implications of data interleaving. Additionally, Betley et al. (2025) and Merity et al. (2017) discuss techniques for training and regularizing models, which can benefit from improved data interleaving strategies to enhance learning efficiency and model robustness.

Furthermore, Harris et al. (2020) and Leiter & Eger (2024) explore array processing and pre-execution models, respectively, which can be optimized through effective data interleaving. These studies underscore the importance of adaptive mechanisms in managing computational resources and improving system throughput.

The adaptation of existing models to incorporate dynamic interleaving strategies is also discussed by Phuntsho et al. (2025), who emphasize the need for flexible frameworks that can adjust to varying data and resource conditions. This aligns with our approach of using canary prompts for dynamic control, offering a pathway to more responsive and efficient systems.

Lastly, Imran et al. (2025) and Turner et al. (2025) provide insights into the theoretical underpinnings of model optimization and scheduling, reinforcing the significance of innovative interleaving techniques in advancing computational efficiency.

In summary, while the literature provides a robust foundation for understanding data interleaving and resource optimization, our work introduces a novel application of canary prompts for dynamic control, filling a gap in the current research landscape.

3 Method

This section outlines the methodologies employed in our research, focusing on the dynamic adjustment of data interleaving ratios to enhance model alignment and performance. The approach is detailed in the following subsection.

3.1 Canary-Controlled Safe-Data Interleaving

Our method leverages a dynamic interleaving strategy to balance safe and task-specific data during the training of language models. The core of this approach is the use of benign 'canary' prompts, which serve as neutral indicators of model alignment. The process begins with the creation of a synthetic dataset that includes a mix of task-specific, safe, and canary prompts. Canary prompts are carefully designed to be neutral, providing a reliable measure of the model's alignment without introducing bias.

During the training phase, the model's performance on these canary prompts is periodically evaluated. This evaluation is crucial as it informs the adjustment of the interleaving ratios of the dataset. By monitoring the model's responses to canary prompts, we can detect signs of emergent misalignment. When such misalignment is observed, the interleaving ratio is adjusted to increase the proportion of safe data, thereby suppressing the misalignment while maintaining the model's coherence and task performance.

This dynamic adjustment process ensures that the model remains aligned with desired outcomes, effectively balancing the need for task-specific learning with the imperative of maintaining safety and ethical standards. The method's adaptability allows for continuous refinement of the interleaving strategy, optimizing the model's performance across various tasks while safeguarding against potential misalignments.

4 EXPERIMENTAL SETUP

In this section, we detail the experimental setup used to evaluate the effectiveness of our dynamic interleaving approach for training language models with synthetic datasets. The experiments were designed to compare the performance of our method against fixed-ratio interleaving and other regularization techniques.

4.1 MODEL ARCHITECTURE AND HYPERPARAMETERS

We utilized a simple Long Short-Term Memory (LSTM)-based language model implemented in PyTorch for our experiments. The model was trained on a synthetic dataset that included interleaved task-specific and safe-data prompts. The primary evaluation metric was Canary Prompt Accuracy, which measures the model's ability to accurately predict task-specific prompts while maintaining safety constraints.

Key hyperparameters for the model included: - **Embedding Dimensions**: The size of the word embeddings used in the model. - **Hidden Dimensions**: The number of units in the LSTM's hidden layer, which determines the model's capacity to learn complex patterns. - **Learning Rates**: The step size used during optimization, crucial for balancing convergence speed and stability.

Hyperparameter tuning was conducted with a focus on the number of training epochs. We iterated through values of 10, 20, and 30 epochs to assess the model's performance across different training durations. This approach allowed us to determine the optimal number of epochs required for effective learning and alignment.

The experiments aimed to demonstrate the superiority of our dynamic interleaving approach by comparing it with fixed-ratio interleaving and other regularization techniques. Results indicated that the model achieved nearly 100% validation accuracy within a few epochs, showcasing its ability to learn effectively from the interleaved dataset. Figure 1 illustrates the training and validation loss curves, as well as the validation accuracy across different epoch settings, providing a visual representation of the model's performance over time.

5 RESULTS

In this section, we present the results of our experiments conducted using a synthetic dataset with interleaved task-specific and safe-data prompts. The experiments utilized a simple LSTM-based language model implemented in PyTorch, with key hyperparameters such as embedding dimensions, hidden dimensions, and learning rates carefully selected to optimize performance. The primary evaluation metric was Canary Prompt Accuracy, which served as a measure of the model's ability to correctly predict task-specific prompts while maintaining alignment with safe-data prompts.

5.1 MODEL PERFORMANCE AND HYPERPARAMETER TUNING

The primary focus of our experiments was to evaluate the effectiveness of our dynamic interleaving approach compared to fixed-ratio interleaving and other regularization techniques. To this end, we conducted hyperparameter tuning by varying the number of training epochs, testing values of 10, 20, and 30 epochs to determine the optimal training duration for our model.

The results demonstrated that our model achieved nearly 100% validation accuracy within a few epochs, underscoring the efficacy of the dynamic interleaving approach. This high level of accuracy indicates that the model was able to effectively learn and align with both task-specific and safe-data prompts, achieving a balance that other regularization techniques struggled to maintain. The rapid convergence to near-perfect accuracy suggests that the chosen hyperparameters, particularly the number of epochs, were well-suited to the task, allowing the model to quickly adapt and generalize from the interleaved dataset.

Overall, the results highlight the potential of dynamic interleaving as a robust method for training language models on datasets that require careful balancing of multiple prompt types. The nearly 100% validation accuracy achieved across different epoch settings confirms the model's capability to effectively integrate and respond to diverse prompts, setting a promising precedent for future research in this area.

6 Conclusion

In conclusion, our dynamic canary-controlled approach to data interleaving effectively addresses the challenge of emergent misalignment while preserving model coherence and task performance. By leveraging canary prompts for dynamic adjustments, this method introduces an innovative mechanism for sustaining model alignment. The promising results underscore the potential of this approach, paving the way for future research to investigate its scalability and applicability to more complex datasets and architectures. This advancement not only enhances the robustness of model alignment strategies but also opens new avenues for optimizing machine learning systems in increasingly intricate environments.

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